4 **(a)** Two light identical springs are joined in series and a 120 g bar magnet is hung from the lower end as shown in Figure 4.1. The effective spring constant of the combined springs is 11 N m^{-1}.

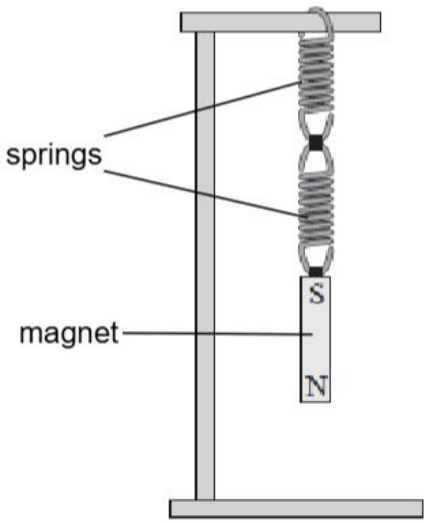


Figure 4.1

The bar magnet is displaced a small distance vertically downward from its equilibrium position and released. It then oscillates in simple harmonic motion.

(i) By considering the forces on the magnet at its equilibrium position and also at a small displacement x downward from its equilibrium position, show that, when released, the magnet's acceleration a is related to its displacement x by the equation:

$$a = -92x$$

Explain your working.

[3]

		L2	**Take direction downward as positive.** **At equilibrium position,** **Upward spring force = Downward weight** **So, Net force = 0.** **At displacement x (downward) below the equilibrium position, spring force increases while weight is unchanged, so** **Upward spring force > Downward weight.** **Net force = -(increase in spring force) = -kx** (negative direction since Net force acts upwards) **By Newton's 2nd law of motion,** **Net force = ma** **-kx = ma** $\mathbf{a = -\frac{k}{m}x}$ $\mathbf{a = -\frac{11}{120\times10^{-3}}x}$ $a = -91.667\,x$ $a = -92\,x$ (shown) OR **At Equilibrium position, no net force.** **Upward Spring force = Downward Weight** **$k\,x_o = mg$ where x_o: spring extension at Equilibrium —(1)** **Take direction downward as positive.** **At displacement x (downward) below the equilibrium position, spring extension is now ($x_o + x$), and net force is non-zero.** **Net force = - (New Spring force) + Weight** **$= -k(x_o + x) + mg$ —--(2)** **Sub (1) into (2):** **Net force = $-k(x_o + x) + kx_o = -kx$** **By Newton's 2nd law,** Net force = ma **-kx = ma** $\mathbf{a = -\frac{11}{120\times10^{-3}}x}$ $a = -91.667\,x$ $a = -92\,x$ (shown) ***Marks scheme:*** ***1 mark: Apply Hooke's law for Spring force.*** ***1 mark: Sign convention (show clearly how "-" sign appears).*** ***1 mark: Apply Newton's 2nd law.***	**M1** **M1** **M1**
		(ii)	Using the expression in **(a)(i)**, calculate the angular frequency of oscillations of the magnet.	

			angular frequency = ...rad s^{-1}	[2]
		L1	a = − 92 x **Comparing with the SHM equation, $a = -\omega^2 x$,** **$\omega^2 = 92$** **M1** $\omega = \sqrt{92}$ **ω = 9.5917 = 9.6 rad s^{-1} (or 9.59 rad s^{-1})** **A1** *Do NOT accept the following since candidates are expected to derive from expression in (a)(i).* $\omega = \sqrt{\frac{k}{m}}$	
		(iii)	The amplitude of the oscillations of the magnet is 3.2 cm. The magnet first passes through the equilibrium position at time t = 0. Calculate the magnitude of the displacement of the magnet from its equilibrium position when t = 0.40 s.	
			displacement = ...cm	[2]
		L2	Take downward as positive direction, and, since x = 0 at t = 0, Use $x = -x_0 \sin(\omega t)$ **$x = -$ (3.2 cm) sin (9.5917 × 0.40)** ***Use Radians mode! Allow EFC from (a)(ii).*** **M1** x = + 2.0494 cm Magnitude of displacement is **2.0 cm (or 2.05 cm)** **A1**	
	(b)	Two identical systems shown in Figure 4.1 are now suspended with the North pole of their magnets located inside a coil of wire. The two coils are connected together with conducting wires as shown in Figure 4.2.		

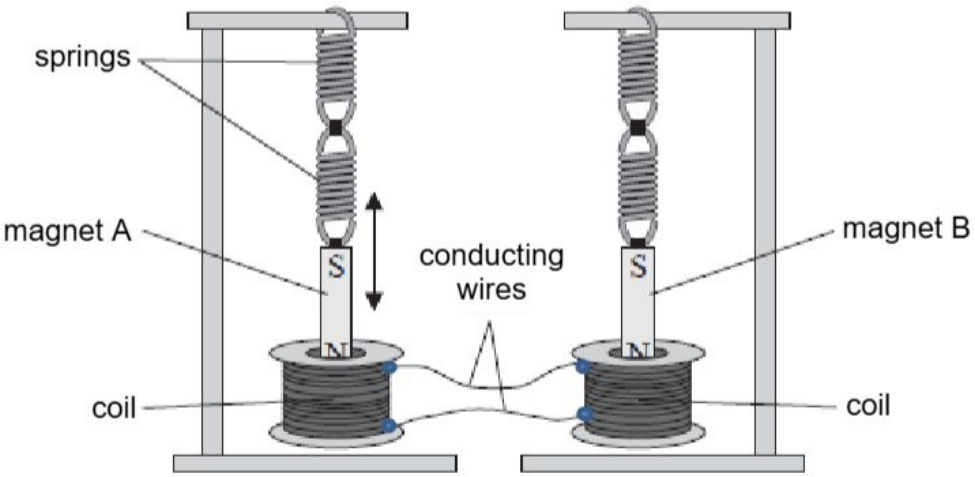


Figure 4.2

Magnet A is displaced so that it oscillates vertically. The North pole of magnet A moves into and out of the coil as it oscillates. As this motion continues, magnet B starts to oscillate. The amplitude of the oscillations of magnet B increases over time.

Explain why magnet B starts to oscillate and why the amplitude of oscillations increases.

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..... [4]

L3 **As the North-pole of magnet A moves into and out of the left coil, the coil experiences alternating increase and decrease in magnetic flux density and hence changes in magnetic flux linkage Φ.** B1

By Faraday's law, this induces an electromotive force (e.m.f.) in the coil (of magnitude proportional to the rate of change of Φ).

By Lenz's law, the direction of the induced e.m.f. reverses periodically. B1

This causes an alternating current to flow in both coils (since the coils are

		connected to each other in a closed circuit). **The alternating current produces a magnetic field in the right coil (below magnet B) that reverses direction periodically, and thus attracts and repels magnet B periodically,** driving it into forced oscillations. **Since both spring-magnet systems have the same natural frequency of oscillations, resonance occurs** and thus magnet B oscillates with increasing amplitude.	**B1** **B1**

[Total: 11]